



Vec-LUT: Vector Table Lookup for Parallel Ultra-Low-Bit LLM Inference on Edge Devices

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Scaling Down Bits for On-Device LLM

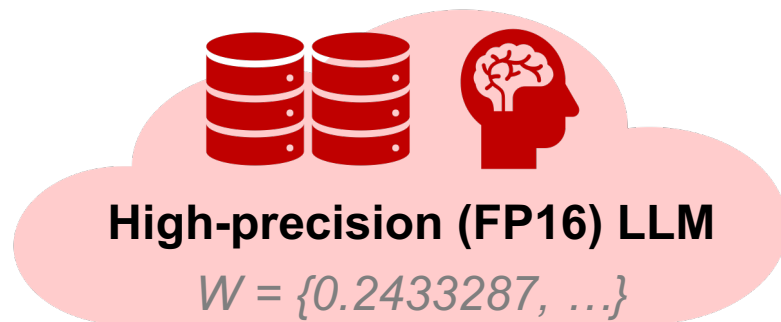
On-device / cloud: a trade-off of intelligence vs. cost / speed /...

- Cloud AI: slow and expensive; most intelligent for general use.
- On-device AI: fast and cheap; sufficient for down-stream tasks [1].

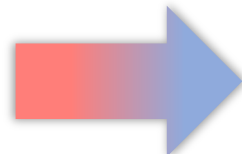


Low-bit quantization: a *de facto* standard for on-device LLMs.

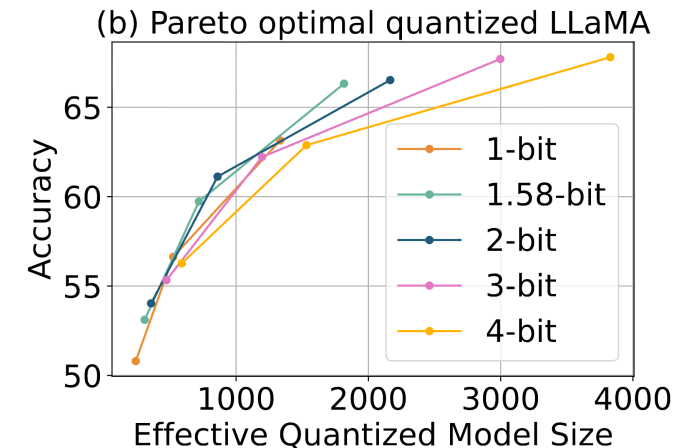
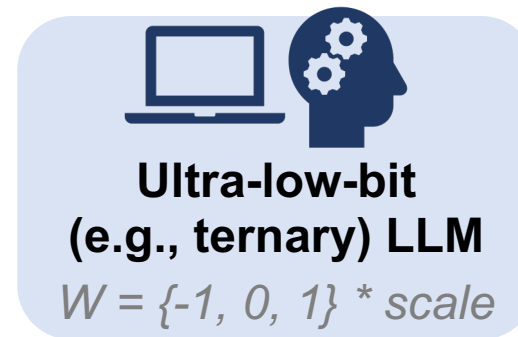
- **Ultra-low-bit (esp. ternary) LLMs demonstrate optimal trade-offs [2, 3].**



Quantize



*GPTQ, AWQ,
BitDistiller, ...*



[1] Belcak P., et al. Small Language Models are the Future of Agentic AI.” 2025, **NVIDIA, Position Paper**.

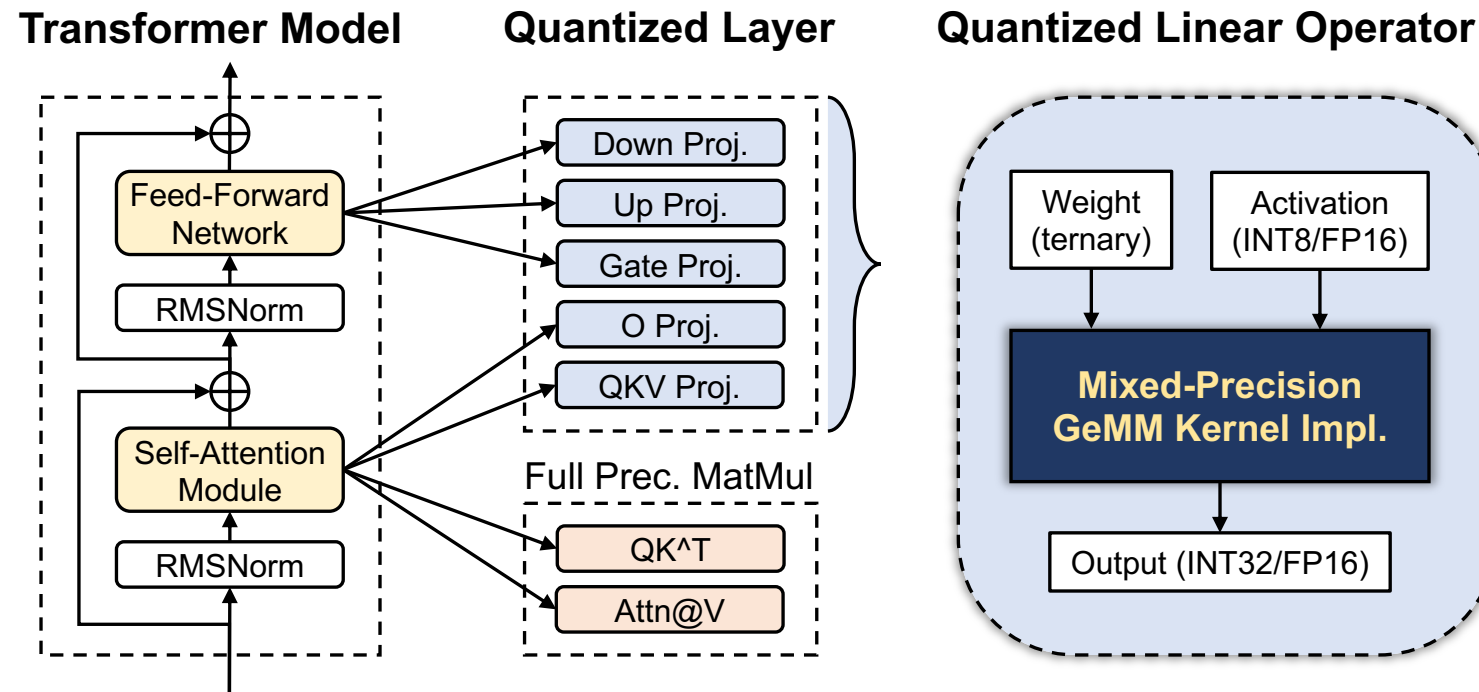
[2] Liu Z., et al. “ParetoQ: Improving Scaling Laws in Extremely Low-Bit LLM Quantization.” **NeurIPS 2025, Meta AI**.

[3] Huang Z., et al. “TENET: An Efficient Sparsity-Aware LUT-Centric Architecture for Ternary LLM Inference on Edge.” 2025, **Microsoft**.

Can Lower Bits Provide Higher Speed?

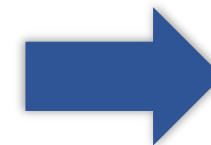
Not always! Only with proper inference kernel design.

- Quantized LLM inference is dominated by *mixed-precision* GeMM (*mpGeMM*).
- **Common hardware (CPU/GPU/...) does not natively support mpGeMM.**
 - E.g., 1.58-bit weight multiplying 16-bit activation.



*How to efficiently implement
mpGeMM without native support?
—— Lookup Table (LUT)*

Deployment

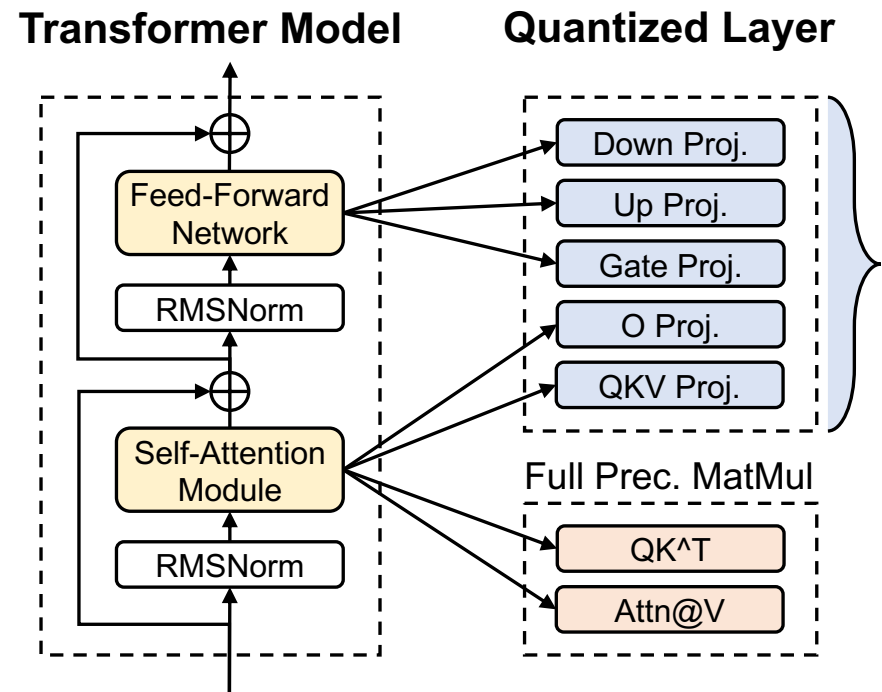


From Dequant. to Lookup Table (LUT)

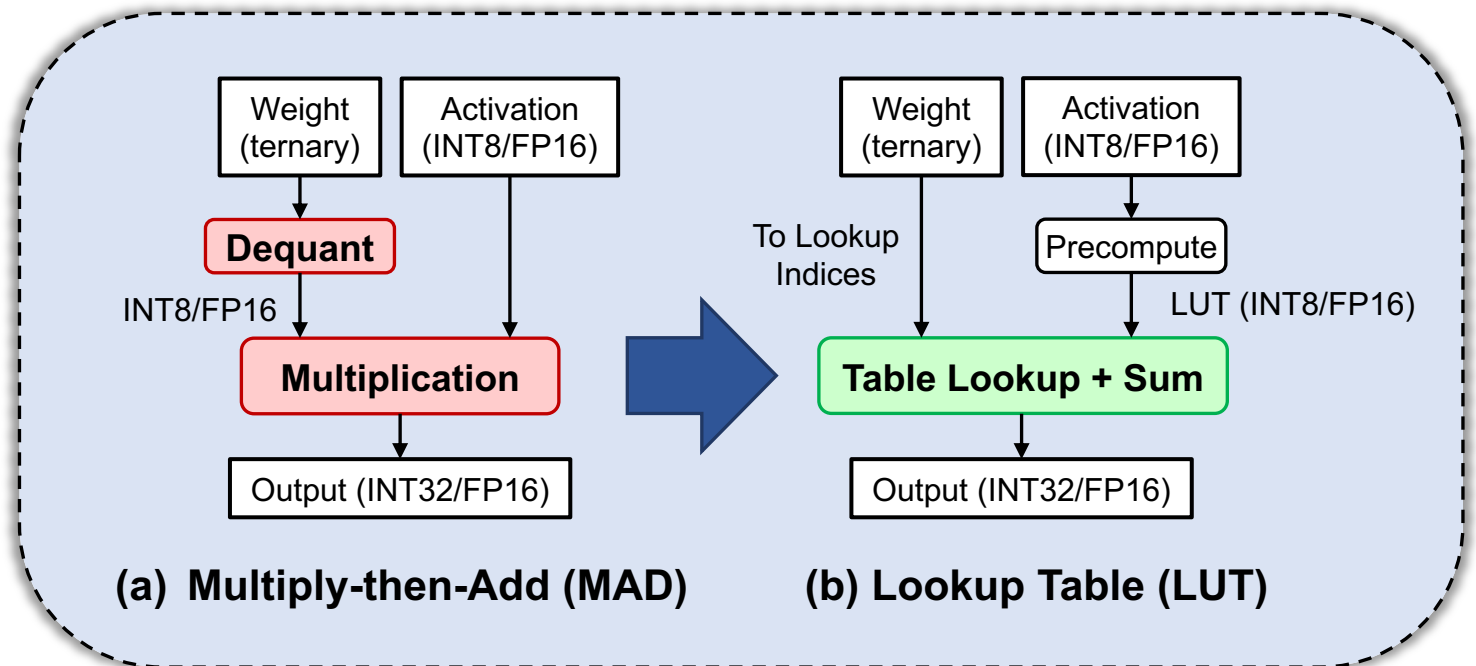
Dequantization and multiplication costs vanish low-bit's gain.

Lookup table (LUT)-based mpGeMM is dequantization-free.

- This is essential for achieving actual speedup from lower bits.*



MAD (dequant.) vs. LUT-based mpGeMM Kernel Impl.



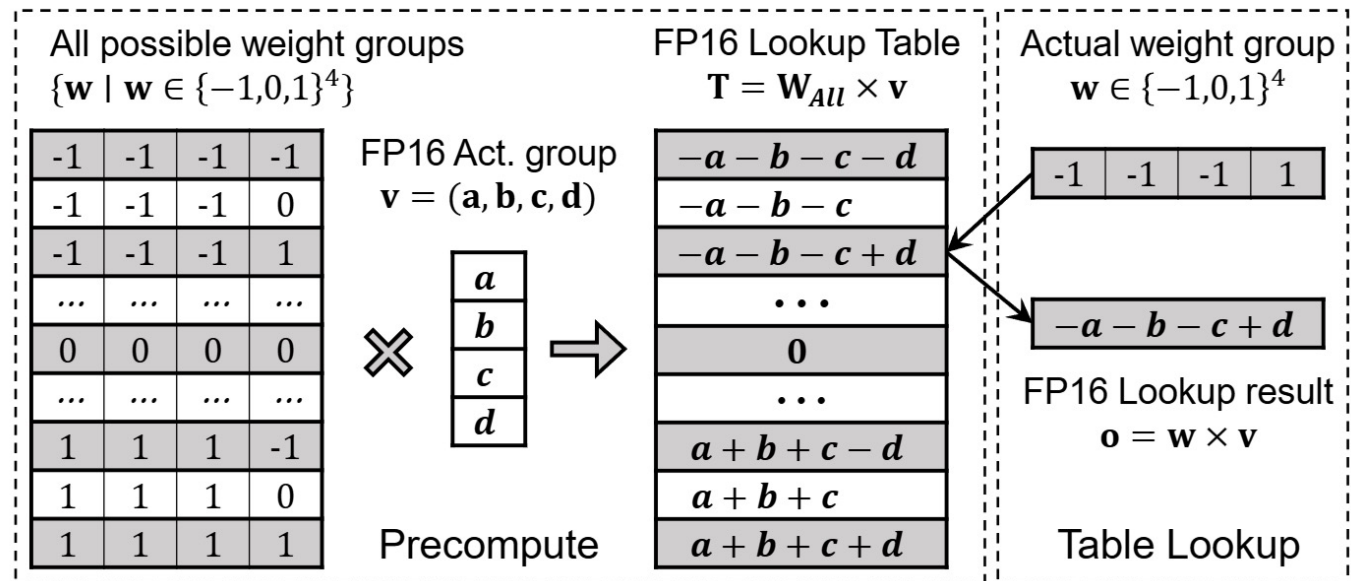
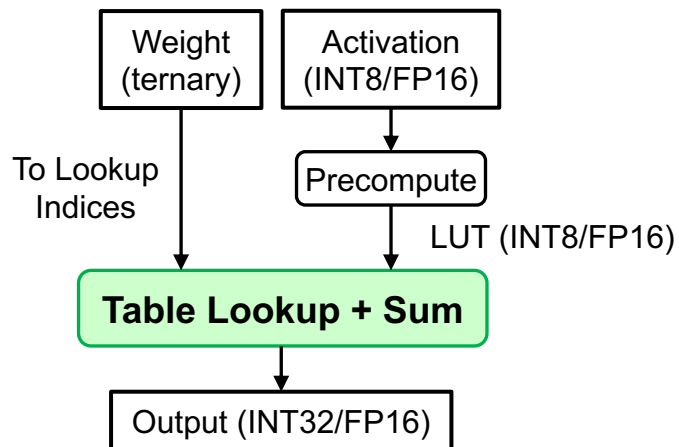
Preliminary: How Does LUT Work?

LUT-based mpGeMM bypasses dequantization by:

1. Enumerating weight patterns (e.g., $\{-1, 0, -1, 1\}$) as *lookup indices*.
2. Precomputing {activation * weight} results as LUT elements.
3. Iterating the actual weight and lookup precomputed results (reused).

Gain: Limited low-bit weight patterns (e.g., 3^4) $\rightarrow$ LUT reused many times.

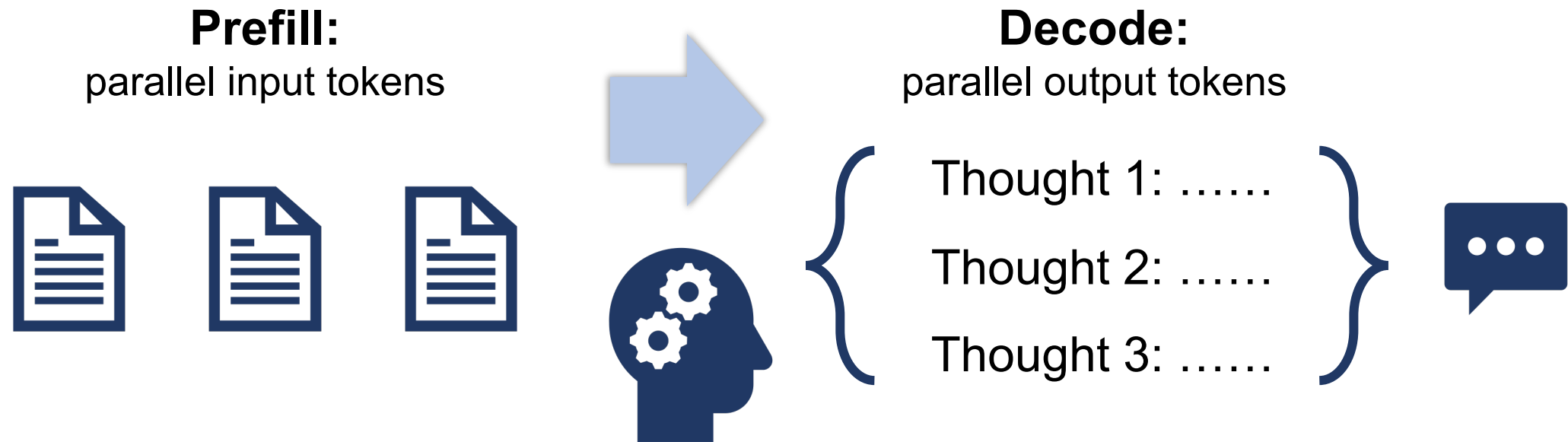
LUT-based mpGeMM Kernel



Inefficiency of LUT in Parallel Inference

Parallel inference: an important part of LLM-based applications.

- Definition: multiple tokens processed together instead of one-by-one.
- **Examples: prefill (medium / large context), parallel / speculative decoding.**



Inefficiency of LUT in Parallel Inference

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Existing LUT-based kernels behave poorly in such scenarios.

- Existing: memory bandwidth util. <40% (even slower than non-LUT kernels).
- **Root cause: the repetitive 1→1 lookup (scalar LUT) for multiple tokens.**

Missing opportunity: index sharing for multiple-token lookups.

- The indices, converted from weights, are **naturally shareable** by tokens.
- **We propose vector LUT to utilize this.**

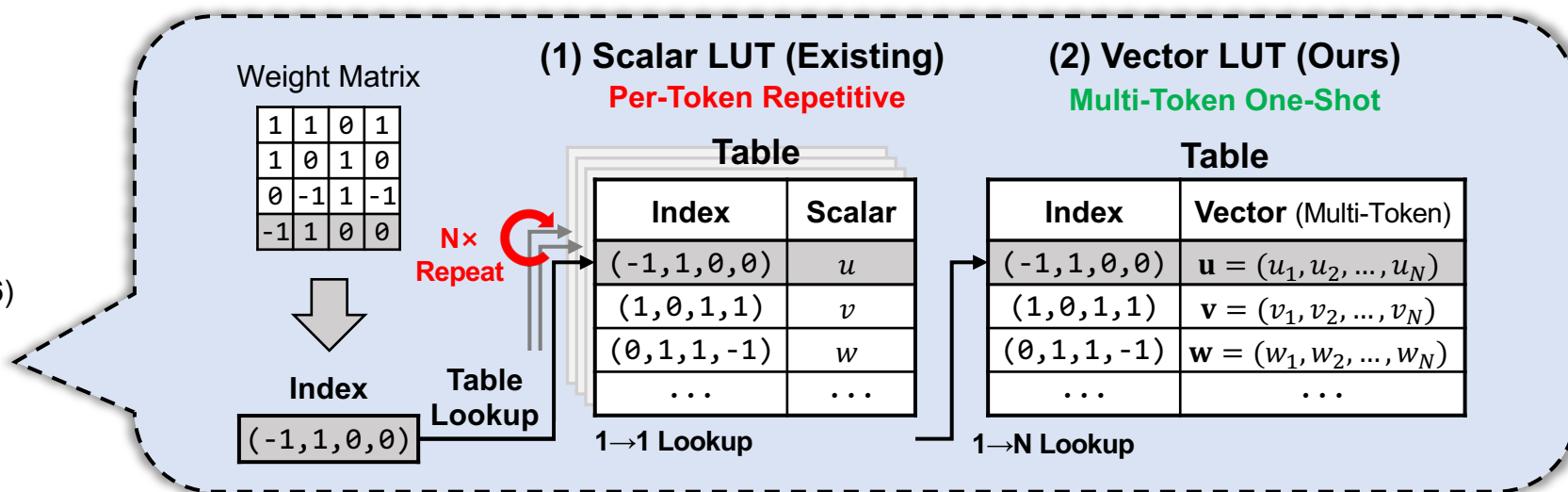
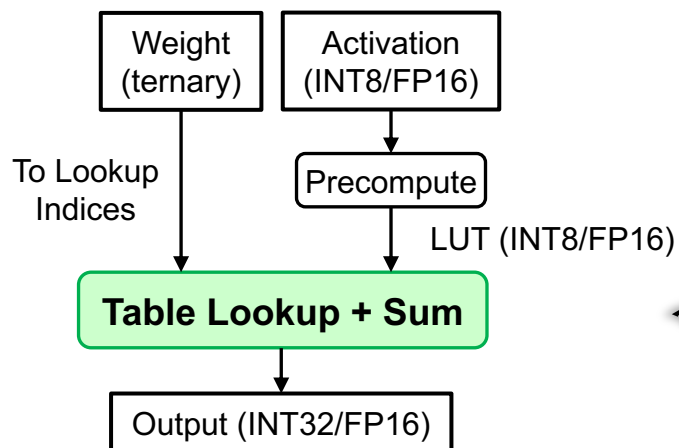
From Scalar LUT to Vector LUT

Vector LUT: $1 \rightarrow N$ lookup with a unified table, sharing one index.

- Main difference: unified table with vector elements, not scalar.
- Primary gain: reducing lookup operations by $N \times$.
- No reliance on hardware lookup instructions, more flexible.

Technical challenges: memory bandwidth and cache capacity.

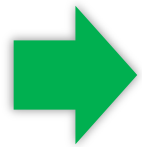
LUT-based mpGeMM Kernel



Vec-LUT: Challenges and Techniques

How to design tensor layouts for memory bandwidth efficiency?

- All tensor layout should match this paradigm for contiguous memory access.
- Non-contiguous memory access can degrade performance by up to 12×.

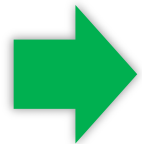


Vector LUT-Centric Tensor Layout

Ensuring contiguous memory access via offline & runtime reorganization.

How to design LUT access to avoid cache thrashing?

- LUT size expanded by N× (e.g., 100s of MiB), exceeding typical cache size.
- Table lookup is inherently **random**, which might lead to a poor cache hit rate.



Cache-Aware Streamed Lookup

Streamlines the precompute-lookup process in tiles within cache capacity.

vlut.cpp: End-to-End Implementation

We implement and integrate Vec-LUT into llama.cpp [1].

- Supports a wide range of edge devices with the ubiquitous CPU backend.
- Pure C/C++ with no external dependencies, built easily on any devices.
- **Source code:** <https://github.com/OpenBitSys/vlut.cpp> .

vlut.cpp is part of OpenBitSys. Other awesome projects (by [Dayou](#)):

- BitDistiller (ACL 2024): QAT with self-distillation for ultra-low-bit LLMs.
- BitDecoding (HPCA 2026): LLM decoding with low-bit KV cache.
- **Follow us on GitHub:** <https://github.com/OpenBitSys> .



[1] <https://github.com/ggml-org/llama.cpp>

On-Device Evaluation & Takeaways

We evaluate Vec-LUT across 5 devices and 3 models.

- Devices (ARM/x86): PC, laptop, smartphone, SBC (Pi), CPU server (AWS).
- Model parameters (different backbones): 1B, 3B, 8B.

Takeaways:

- **Fast:** up to $4.2\times$ (I1, BPW=1.6) and $2.6\times$ (I2, BPW=2.0) E2E speedup.
- **Compact:** 1.6 bits per weight (BPW) packing w.o. strict shape requirements.
- **Generic:** supports mixed prefilling and decoding (e.g., continuous batching).
- **Energy-efficient:** up to $2.1\times$ reduction of energy cost per token.
- **Competitive to NPU:** $1.1\times$ (2 cores) on Snapdragon 8 Elite vs. llama.cpp.

[1] <https://github.com/ggml-org/llama.cpp>

Parallel Decoding: 3× E2E Speedup

Configuration: Llama3 8B decoding on a single CPU core.

vlut.cpp (left) vs. llama.cpp (right) generating 32 sequences in parallel (3x faster).

main: generating 32 sequences with Llama3-8B-1.58-100B-tokens/ggml-model-T1_V_2.gguf

0 I believe that you will find the following information useful. The first is the
1 I believe that that the most important task of the future is to make the
2 I believe the story of the Great Flood is one of the most important stories
3 I believe that the best way to make a positive change is to educate our
4 I believe that the results of the study will be of great interest to the
5 I believe that you are a man of good sense and a good judge of
6 I believe that the world will be better off with the inclusion of the women
7 I believe that this is the best way to present the information, students will
8 I believe that the most important thing to do is to have a good time
9 I believe that the most important thing is to understand that we are born into
10 I believe that the most important thing is to develop a mindset that is willing
11 I believe that you are a very special person. You have a gift to
12 I believe that the world is a better place than it is today, and as
13 I believe that the use of the word
14 I believe that the world is a better place than it is today, and as
15 I believe that the best way to learn about the world is to travel there
16 I believe that the world is a better place than it is today, and as
17 I believe that the world is a better place than it is today, and as
18 I believe that the world is a better place than it is today, and as
19 I believe that you are a great person and I am here to share my
20 I believe that there is a lot of confusion about the differences between the vari
21 I believe that you can get the answer of: what is the difference between
22 I believe that this may be the best way to ensure the safety of our
23 I believe that the next decade will be one of the most exciting and important
24 I believe that the best way to reduce the number of people who die in
25 I believe that the first four days of the week are the most important,
26 I believe that the most important thing for me to do is to help my
27 I believe that this is a very significant discovery, as it could pave the
28 I believe that you, in fact, I know you, that you are
29 I believe that the most important thing in the world is to make the world
30 I believe that this is a very important subject to discuss. I believe it
31 I believe that the world is a better place than it was a decade ago

main: decoded 416 tokens in 29.88 s, speed: 13.92 t/s

llama_perf_sampler_print:	sampling time =	33.93 ms /	448 runs (	0.08 ms p
llama_perf_context_print:	load time =	1817.37 ms		
llama_perf_context_print:	prompt eval time =	30881.01 ms /	419 tokens (	73.70 ms p
llama_perf_context_print:	eval time =	0.00 ms /	1 runs (	0.00 ms p
llama_perf_context_print:	total time =	31701.29 ms /	420 tokens	

main: generating 32 sequences with Llama3-8B-1.58-100B-tokens/ggml-model-TQ1_0.gguf

0 I believe that we have to do something about this, - I believe
1 I believe that I am not the only one in this position. If you
2 I believe that they are the worst things to eat in the world, eat
3 I believe in the importance of teaching students to be responsible global citize
4 I believe that this is the first time I have seen a book in which
5 I believe that the most important factors in the decision to become a parent ar
6 I believe that the future of the world's energy supply will be based on
7 I believe that I have a good understanding of the topic I am writing about
8 I believe that I am a good person. I think most people believe that
9 I believe that this was the case, I have been told that this was
10 I believe that the best way to teach the children is to teach them in
11 I believe that the best way to understand and appreciate the history of a place
12 I believe that the best way to understand and appreciate the history of a place
13 I believe that the best way to understand and appreciate the history of a place
14 I believe that the best way to understand and appreciate the history of a place
15 I believe that the only way to keep the peace is to disarm. And
16 I believe that the only way to keep the peace is to disarm. And
17 I believe that the only way to keep the peace is to disarm. And
18 I believe that the only way to keep the peace is to disarm. And
19 I believe in the following: 1. The universe is eternal and infinite and
20 I believe that that the greatest contribution to the development of the human r
21 I believe that I can help you in this matter, I will be glad
22 I believe that I have a gift for writing. I believe I have a
23 I believe that I can learn to use the internet more effectively for research an
24 I believe that I shall not live to see the downfall of the Nazi regime
25 I believe that I was the first to discover the existence of the Great Lakes
26 I believe that the United States has a great future. I believe this because
27 I believe that they are a great way to learn and they are a great
28 I believe that I am a good person. But I don't know how
29 I believe the story of the first Thanksgiving is a myth, story of the
30 I believe that you will be able to find the answer to your question,
31 I believe that it is only a matter of time before the next pandemic hits

main: decoded 416 tokens in 91.05 s, speed: 4.57 t/s

llama_perf_sampler_print:	sampling time =	34.10 ms /	448 runs (	0.08 ms
llama_perf_context_print:	load time =	1176.35 ms		
llama_perf_context_print:	prompt eval time =	91480.70 ms /	419 tokens (	218.33 ms
llama_perf_context_print:	eval time =	0.00 ms /	1 runs (	0.00 ms
llama_perf_context_print:	total time =	92222.40 ms /	420 tokens	

When to Use Ultra-Low-Bit?

- Domain-specific tasks (not requiring world knowledge).
- Enough training data available (for QAT).
- Memory usage / inference speed is the priority.

Toward Agentic / Physical AI

We are working on:

- **Quantization-aware training and inference systems.**
 - 2-bit training and serving: in collab. with Hisense, releasing soon.
 - More complex tasks (e.g., agentic): WIP.
- **Self-evolving physical agents.** Recent preprints:
 - OxyGen [1]: accelerating multi-task VLA inference on robots.
 - ActProbe [2]: detecting robot policy failures ahead of time.
 - EmbodiSkill [3]: skill reflection and refinement after deployment.
 - MemCompiler [4]: state-conditioned agent memory compilation.

Discussion / collaboration / join us :)

[1] Li, X., et al. "OxyGen: Unified KV Cache Management for VLA Inference under Multi-Task Parallelism." 2026.

[2] Huang, B., et al. "ActProbe: Action-Space Probe for Early Failure Detection of Generative Robot Policies." 2026.

[3] Ju, R., et al. "EmbodiSkill: Skill-Aware Reflection for Self-Evolving Embodied Agents." 2026.

[4] Ding, X., et al. "MemCompiler: Compile, Don't Inject--State-Conditioned Memory for Embodied Agents." 2026.

Thanks!

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